

Junjie (Caspar) Chen

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EXPERIENCE

Taskrabbit, Inc.

Marketplace Platform — IKEA Group · terabyte-scale Snowflake data warehouse

San Francisco, CA

Data Engineer II

Nov 2022 – Present

- **Architected Ask Taskrabbit, a company-wide Claude-based text-to-SQL analytics product**, enabling employees to query governed company metrics in natural language and receive consistent, dashboard-aligned answers at approved grains and dimensions. Connected Claude to dbt via MCP and to Snowflake through a least-privilege service role scoped to approved databases and schemas.
- **Built the dbt Semantic Layer and LLM metric guardrails as the canonical KPI contract**, standardizing metric names, grain/dimensions, aggregation-safe ratio formulas (IBR, AOV, Take Rate, QFR), and LY/WoW comparison logic so AI-generated SQL follows the same business definitions used in production dashboards.
- **Architect and operate Taskrabbit's terabyte-scale Snowflake data platform**, owning multi-layer warehouse design (staging → intermediate → marts) and dbt models supporting Finance, Marketing, Operations, and executive reporting.
- **Led greenfield data platform buildout for the Dolly acquisition**, designing the dbt repository, transformation architecture, and staging/intermediate/mart layers to integrate acquired data assets into Taskrabbit's unified warehouse.
- **Reduced Snowflake compute cost by 30%+** through query profiling, incremental modeling, partition pruning, clustering optimization, and reusable SQL/Jinja patterns.
- **Built and operated production Airflow (Astronomer) pipelines** with dynamic task mapping, cross-DAG dependencies, and resilient retry logic for daily workflows processing millions of marketplace events.
- **Designed multi-source ingestion pipelines** across REST APIs (Stripe, Fixer, Segment, Attom), S3, and internal microservices using Fivetran and custom Python connectors, enforcing schema contracts across 160+ production tables.
- **Designed Snowflake release architecture** across isolated dev, staging, and prod environments with role-based access controls, establishing a controlled dev → staging → prod release process. Integrated dbt Cloud CI with modified+ selection and --fail-fast to execute changed models and downstream dependencies in staging before promotion, and automated scheduled-job releases through Jenkins, Ansible, and cron.
- **Drove engineering standards across 5+ data repositories** through routine code review, SQLFluff quality gates, and test-driven development in Jira spike tickets, validating design assumptions, edge cases, and failure modes before production implementation.
- **Designed and operated event-driven Snowflake Streams/Tasks pipelines**; following a Snowflake outage, detected silently suspended tasks and stale streams, reconstructed missed execution windows with custom SQL based on task logic, backfilled affected data, reactivated task dependencies, and validated downstream freshness and completeness.
- **Planned and executed historical backfills** during Snowflake database-object migrations, reconciling legacy data into replacement objects and validating row counts, downstream model outputs, and business metrics before cutover.
- **Resolved a \$4.4M IKEA billing variance to \$14K** across four European markets (UK, ES, PT, FR) by tracing systemic tracking and calculation errors across warehouse models and accounting workflows.
- **Built automated Stripe-to-CSI revenue reconciliation** in Snowflake and Python, modeling charges, transfers, refunds, and platform earnings to detect transaction anomalies and improve end-to-end financial integrity.
- **Codified Snowflake infrastructure and governance in Terraform**, including role/grant management, dynamic data masking, column tagging, automated tests, and quality alerting, reducing mean incident response time by 40%.

Data Engineer I

Sep 2021 – Oct 2022

- **Led migration of legacy data models** to Fivetran, dbt, Snowflake, and Looker, reducing ETL runtime from 8 hours to under 30 minutes while maintaining downstream data quality.
- **Owned production ETL reliability** by triaging urgent pipeline failures, restoring data availability, and maintaining SLA compliance for analytics and reporting consumers.
- **Developed dbt models and Airflow DAGs** for marketing attribution and operations reporting, enabling campaign performance tracking and KPI visibility across business teams.

Apple (via Apex Systems)

Sunnyvale, CA

Data Analyst Lead

Dec 2020 – Sep 2021

- **Built geospatial ingestion and transformation pipelines** using SQL, Scala, QGIS, PostgreSQL, and Hive/Hadoop to integrate multi-source mapping datasets into Apple Maps data systems.
- **Owned data validation and remediation workflows** for OpenStreetMap ingestion across 8 countries, developing quality checks and transformation logic to resolve geographic data inconsistencies before downstream consumption.

EDUCATION

Columbia University

M.A. in Statistics

New York, NY

Sep. 2018 – Dec. 2019

Coursework: Data Mining, Databases, Data Science, Time Series Analysis, Nonparametric Methods

University of Wisconsin–Madison

B.S. Applied Mathematics, Actuarial Science

Sep. 2014 – May. 2018

SKILLS & CERTIFICATIONS

Languages & Engineering: Python · SQL · Scala

Data Platform: Snowflake · dbt · Airflow (Astronomer) · Fivetran · Segment · Kafka

Cloud & Infrastructure: AWS · S3 · Terraform

CI/CD & Engineering: dbt Cloud CI · Jenkins · Ansible · GitHub · TDD · Dev/Staging/Prod Releases

Data Modeling & Reliability: Dimensional Modeling · Star Schema · ELT · Data Quality · Data Observability · RBAC

BI & Analytics: Sigma · Looker · Tableau · Mixpanel

AI & Semantic Layer: Claude · dbt MCP · Text-to-SQL · dbt Semantic Layer · LLM Guardrails

Certifications: AWS Certified Cloud Practitioner · Dbt Certified Developer · Astronomer Certified DAG Authoring for Apache Airflow